Foundational LLMs - Easy Notes for Study & Explanation

1. What is an LLM?

A Large Language Model (LLM) is an AI system that predicts the next word or token in a sentence. By learning from huge amounts of text, it can answer questions, write content, summarize text, and generate code.

2. Transformer Architecture

Transformers are the technology behind modern AI models. They process many words at the same time instead of one-by-one, making them faster and more powerful.

3. Tokenization and Embeddings

Tokenization breaks text into small pieces called tokens. Embeddings convert words into numbers so computers can understand relationships between words.

4. Self-Attention

Self-attention helps the model understand which words are related. Example: In 'The tiger drank water because it was thirsty', the model learns that 'it' refers to 'tiger'.

5. Multi-Head Attention

Different attention heads look at different relationships such as grammar, meaning, and context. Together they improve understanding.

6. Encoder and Decoder

Encoder = understands input. Decoder = generates output. GPT models mainly use decoder-only architecture.

7. Training

The model reads huge datasets, predicts missing or next words, checks mistakes, and improves itself repeatedly.

8. GPT Evolution

GPT-1 introduced transformer-based language generation. GPT-2 became larger. GPT-3 scaled to 175 billion parameters. GPT-4 added stronger reasoning and multimodal abilities.

9. BERT

BERT focuses on understanding language rather than generating text. It is excellent for classification and question answering.

10. Fine-Tuning

Fine-tuning specializes a general model for a specific task such as law, healthcare, customer support, or coding.

11. RLHF

Reinforcement Learning from Human Feedback improves model behavior using human preferences, making responses safer and more helpful.

12. LoRA and PEFT

These methods make fine-tuning cheaper by training only a small number of additional parameters instead of the whole model.

13. Prompt Engineering

Prompt engineering means writing clear instructions. Better prompts usually produce better results.

14. Sampling Techniques

Temperature controls creativity. Low temperature = predictable answers. High temperature = creative answers.

15. Inference Optimization

Methods such as quantization and distillation make models faster and cheaper while maintaining acceptable quality.

16. Gemini

Gemini is a multimodal AI model that can work with text, images, audio, and video. It supports very large context windows.

Easy Real-Life Analogy

Think of an LLM as a very intelligent student who has read millions of books. Transformers are the student's brain, attention is focus, training is studying, and fine-tuning is specialization.

Quick Revision Questions

1. What is the main job of an LLM?
2. Why are Transformers important?
3. What is tokenization?
4. Explain self-attention with an example.
5. What is the difference between encoder and decoder?
6. What is fine-tuning?
7. What is RLHF?
8. Why is LoRA useful?
9. What is prompt engineering?
10. What does temperature control?